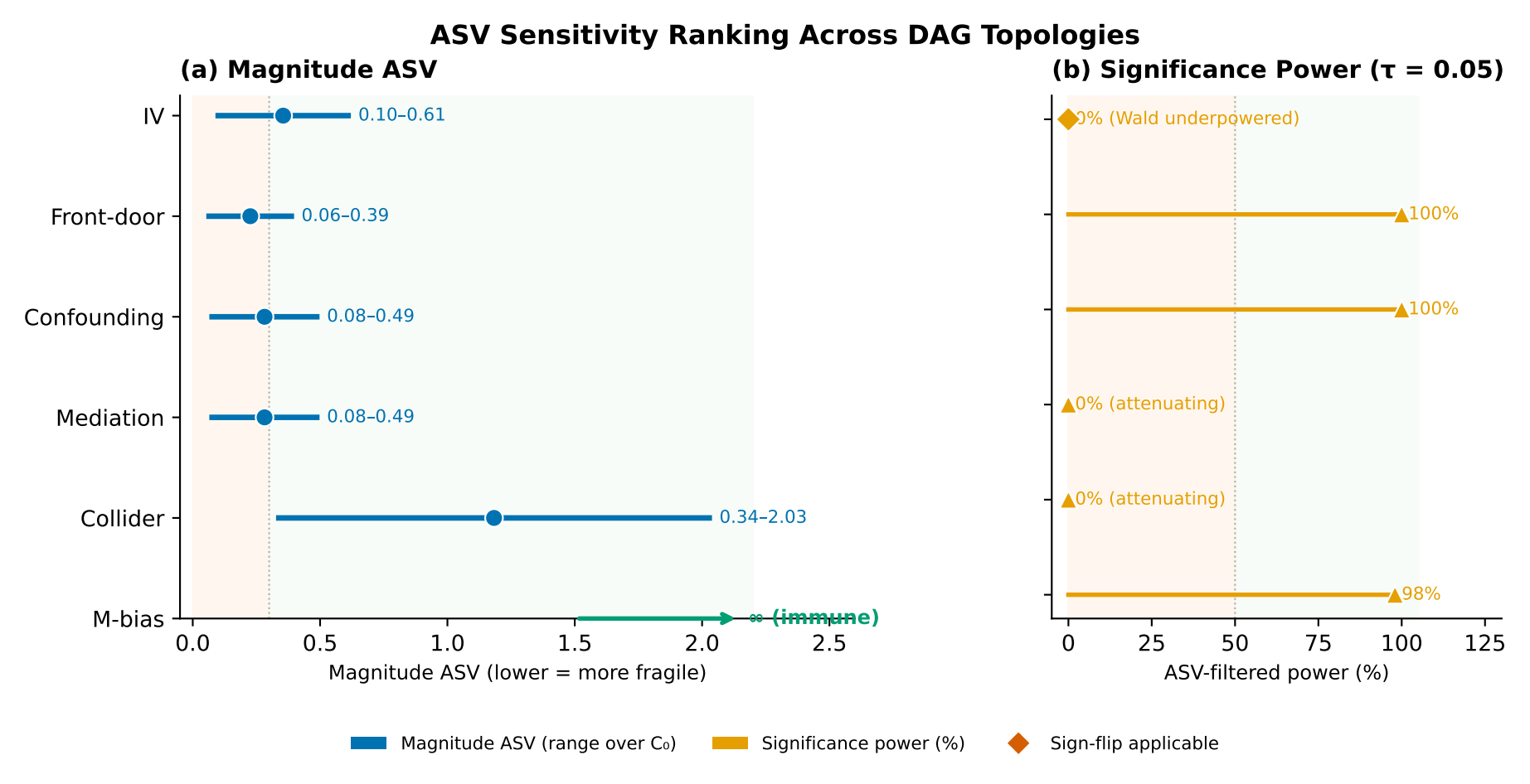
ASV Sensitivity Ranking Across DAG Topologies
(a) Magnitude ASV
(b) Significance Power (τ = 0.05)
IV
Front-door
Confounding
Mediation
Collider
M-bias
0.10–0.61
0.06–0.39
0.08–0.49
0.08–0.49
0.34–2.03
∞ (immune)
0.0
0.5
1.0
1.5
2.0
2.5
Magnitude ASV (lower = more fragile)
0% (Wald underpowered)
100%
100%
0% (attenuating)
0% (attenuating)
98%
0
25
50
75
100
125
ASV-filtered power (%)
Magnitude ASV (range over C₀)
Significance power (%)
Sign-flip applicable